IA4- Assignment

Due date: 08/11/2025

Instructions:

1. **Upload your completed assignment in LMS and submit a hard copy to the respective teacher**
2. **Write your assignment on A4 paper and include your name, roll number, register number, and section at the top left corner of the first page.**

1. An ideal 1 μs pulse is fed to an amplifier. Calculate and plot the output waveform under the following conditions, The upper 3 dB frequency is: (i) 10 MHz (ii) 0.1 MHz.

2. A 1 KHz symmetrical square wave of ± 10 V is applied to RC circuit having 1 msec time constant. Calculate and plot the output to the scale for RC configurations as,

(i) High pass circuit

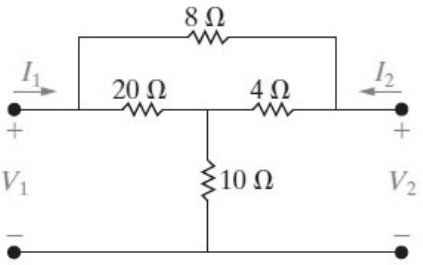
(ii) Low pass circuit

3. A 10 KHz square wave is applied to high pass RC circuit which produces the output with a tilt of 3.8%. Calculate the lower 3-aB frequency of the circuit. If the circuit uses a capacitor of 0.47 μF, determine the value of the resistance.

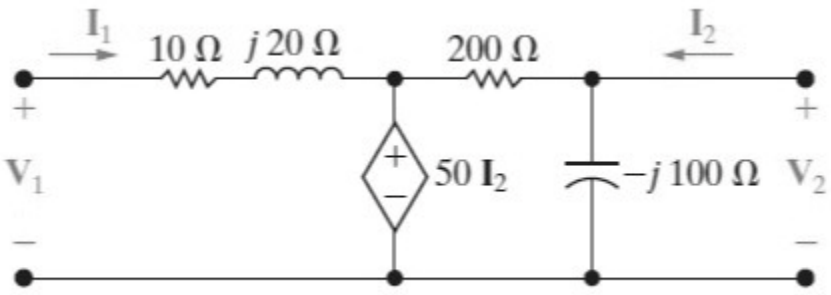
4. Explain why and under what conditions a high-pass RC circuit can function as a differentiator.

5. Prove that a Low-Pass RC Circuit behaves as an Integrator.

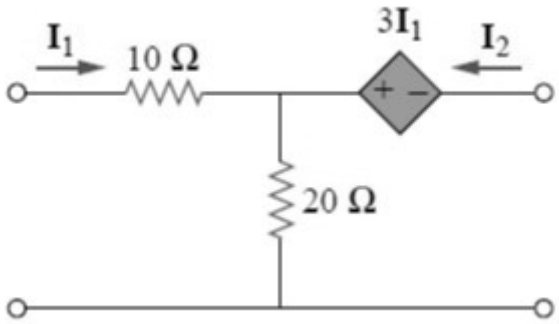
6.Find the y parameters for the circuit shown in Fig.



7. Find the h parameters of the two-port circuit shown in Fig.



8. Find the transmission parameters for the two-port network in Fig.



9. Find [z] and [g] of a two-port network if-

$$[\mathbf{T}] = \begin{bmatrix} 10 & 1.5\ \Omega \\ 2\ \text{S} & 4 \end{bmatrix}$$

10. For the network shown in Fig,, find Z and Y-parameters.

